



Data Article

Thalassemia dataset covering clinical, socioeconomic, and mental health aspects



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ABSTRACT

Thalassemia is an inherited disorder of haemoglobin formation that requires lifelong blood transfusions and iron removal and brings both medical and emotional challenges. This dataset contains detailed information on 617 patients with thalassemia treated at a leading diagnostic centre in Pabna, Bangladesh. Demographic and socioeconomic data were gathered through face-to-face interviews, including age, gender, place of residence (urban or rural), education level, monthly household income, travel time to hospital, and how thalassemia affects school or work life. Clinical details, such as the type of thalassemia, age at diagnosis, whether the spleen was removed, transfusion frequency, the iron chelation plan, and how well patients adhered to it were taken from medical records. Laboratory results from the hospital's central lab include pre transfusion haemoglobin, serum ferritin, red blood cell indices, reticulocyte count, and foetal haemoglobin percentage. Patients also rated their recent mental health as good or bad. Each record is assigned a unique code and stored in a single CSV file with twenty-three variables. The combination of medical, laboratory, social, and psychological information can support many uses, from building machine learning models to predict transfusion

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needs or therapy adherence to studying factors that drive health disparities and designing more patient centred care plans.

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Specifications Table

Subject	Health Sciences, Medical Sciences & Pharmacology
Specific subject area	Thalassemia patient data with demographic, clinical, socioeconomic, and mental health attributes.
Type of data	Raw, Processed
Data collection	Collected from 617 thalassemia patients (March–June 2025) at LABAID LTD., Pabna, Bangladesh, via face-to-face interviews and medical records. Includes demographic, socioeconomic, clinical, lab, and mental health data. Stored as a de-identified CSV with twenty-three variables.
Data source location	LABAID LTD. (Diagnostic), Pabna, Bangladesh (Latitude and Longitude: 24.0081796°N,89.2389675°E)
Data accessibility	Repository name: Mendeley Data Data identification number (DOI): 10.17632/8kcdkxmcjw.1 Direct URL to data: https://data.mendeley.com/datasets/8kcdkxmcjw/1 Instructions for accessing these data: The dataset is licensed under CC BY 4.0 and is freely available for download without registration. Includes one UTF-8 encoded CSV file named Thalassemia_Bangladesh_dataset.csv.
Related research article	None

1. Value of the Data

- This dataset offers a unique, multidimensional view of 617 thalassemia patients by combining clinical, laboratory, demographic, socioeconomic, and mental health data into a single, structured resource.
- This dataset's mix of numerical, ordinal, and categorical features enables machine learning models to predict clinical outcomes like transfusion needs, iron chelation adherence, and mental health, supporting personalized treatment plans.
- Combining self-reported mental health, school/work impact, and clinical measures like hemoglobin and transfusion frequency allows a comprehensive analysis of thalassemia's physical and psychological burdens.
- Researchers, healthcare professionals, and policymakers can use this dataset to explore the relationships between chronic illness management, psychological well-being, and social context, helping to improve patient-centered interventions and public health strategies.

2. Background

Thalassemia is an inherited blood disorder that affects the body's ability to make healthy hemoglobin, leading to ongoing anemia [1]. The World Health Organization (WHO) defines anemia as a blood hemoglobin (Hb) concentration of less than 13 g/dL in male patients and less than 12 g/dL in female patients [2]. People with severe forms of thalassemia often need regular blood transfusions for life, which can cause other health problems like too much iron in the body and damage to organs. It also affects how well they can live day to day. Thalassemia is especially common in places like South Asia, the Middle East, and the Mediterranean, where it's a serious public health issue [3]. Even though treatments like blood transfusions and iron-removal medicines help, managing the disease is still difficult because of limited access to care, challenges with sticking to treatment, and the emotional stress patients often go through. Managing

thalassemia involves more than just medical treatment. Many patients also face challenges like living far from healthcare centers, financial difficulties, and limited education, all of which can make it harder to manage the disease effectively [4,5]. On top of that, the stress of ongoing treatments can take a toll on mental health, leading to anxiety, depression, and a lower quality of life. Although thalassemia is common in many parts of the world, there's still a shortage of rich, well-rounded datasets that combine medical, social, and mental health information. This dataset helps fill that gap by offering a complete picture of the patient experience, making it easier to explore how different factors are connected and to support more personalized, patient-focused care and research.

3. Data Description

The dataset covers 24 different topics, like age, income, treatment history, and how the condition affects daily life, giving a full picture of each person's medical and personal situation. The variables included in the dataset are detailed in Table 1, which lists each variable's name, description, data type, and units or categories where applicable.

4. Experimental Design, Materials and Methods

4.1. Study setting and participants

From March to June 2025, we collected data from thalassemia patients at LABAID LTD. (DIAG-NOSTIC) PABNA, a major healthcare center in Bangladesh that serves both urban and rural communities. Patients of all ages visiting the transfusion clinic were invited to take part in the study

Table 1
Summary of variables in the thalassemia patient dataset.

Variable Name	Description	Data Type	Units / Categories
Patient_ID	Unique identifier for each patient	Integer	N/A
Current_Age	Age at the time of data collection	Numerical	Years
Gender	Gender of the patient	Categorical	Male, Female
Residence_Type	Area type where the patient lives	Categorical	Rural, Urban
Education_Level	Highest education level attained	Categorical	None, Preschool, Primary, Secondary, Graduate, Postgraduate
Monthly_Household_Income	Total family income per month	Numerical	Bangladeshi Taka
Travel_Time_to_Hospital	Time taken to reach the hospital	Numerical	Minutes
School/Work_Impact	Impact on education or job	Ordinal	None, Mild, Moderate, High
Age_at_Diagnosis	Age when thalassemia was first diagnosed	Numerical	Years
Thalassemia_Type	Type of thalassemia	Categorical	Major, Intermedia, Minor
Splenectomy_Status	History of spleen removal	Categorical	Yes, No
Self-Reported_Comorbidities	Other health issues reported	Text	N/A
Pre-Transfusion_Hb	Hemoglobin level before transfusion	Numerical	g/dL
Serum_Ferritin	Ferritin level in blood	Numerical	ng/mL
MCV	Mean corpuscular volume	Numerical	fL
MCH	Mean corpuscular hemoglobin	Numerical	pg
RDW	Red cell distribution width	Numerical	%
Reticulocyte_Count	Reticulocytes percentage	Numerical	%
HbF_Percent	Fetal hemoglobin percentage	Numerical	%
Transfusion_Frequency	Monthly transfusion rate	Numerical	Units/month
Iron_Chelation_Therapy	Iron chelation method	Categorical	DFX, DFP, DFX+DFP, None
Chelation_Adherence	Adherence to chelation therapy	Ordinal	Low, Medium, High, N/A
BMI	Body Mass Index	Numerical	kg/m²
Mental_Health_Status	Self-reported mental health	Categorical	Good, Bad

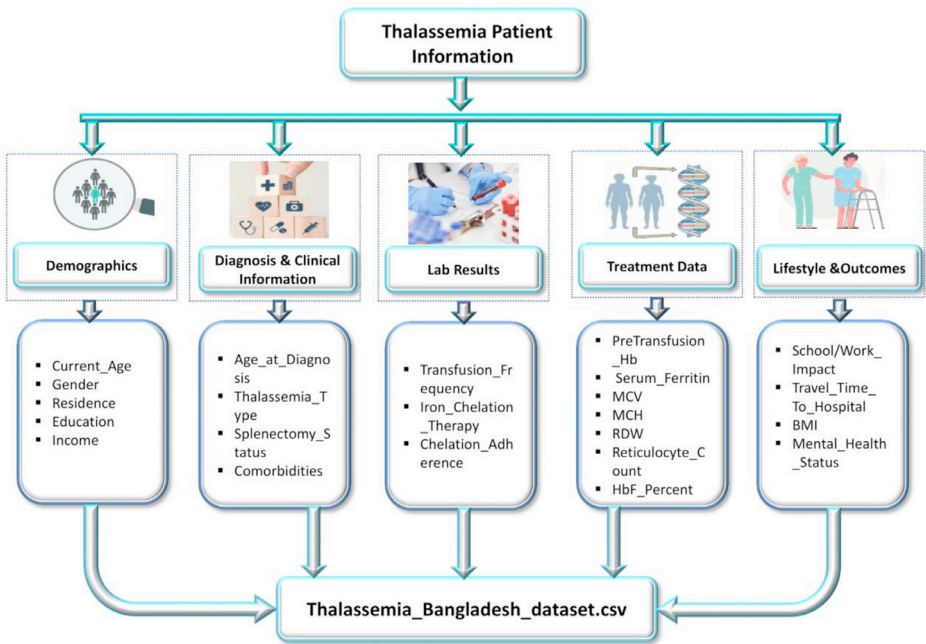


Fig. 1. Structured grouping of thalassemia dataset variables into thematic categories.

if they had confirmed diagnosis of thalassemia (major, intermedia, or minor), attended regular follow-ups, and provided informed consent (or parental consent for minors). Genetic subtyping (e.g., alpha vs. beta thalassemia) was not performed, but given the clinical context and national prevalence, most cases are presumed to be beta-thalassemia. We entered all patient responses directly into a CSV file on our laptop during the interviews. No personal identification information was collected. Patients with unrelated serious health conditions were excluded from the study. After collecting data from 617 patients with different types of thalassemia, the doctors and hospital authorities reviewed and fully validated the dataset.

4.2. Data collection procedures

The data include demographic and socioeconomic details, such as age, gender, residence (urban or rural), education level, and monthly household income: collected through face-to-face questionnaires conducted by trained research assistants. Information on travel time to the hospital and the impact of thalassemia on school or work life was also recorded. Clinical information like thalassemia type, age at diagnosis, splenectomy status, and other health conditions were taken from their medical records, along with treatment details such as how often they receive blood transfusions, what type of iron chelation therapy they use, and how regularly they follow it. Laboratory parameters including pretransfusion hemoglobin, serum ferritin, MCV, MCH, RDW, reticulocyte count, and fetal hemoglobin percentage were collected from the hospital's central laboratory. Mental health status was evaluated through a straightforward self-report question: "How would you rate your mental health over the past few days?" Patients responded with either "Good" or "Bad", providing a quick assessment of their recent psychological wellbeing. This binary indicator was chosen to balance rigor with feasibility in a busy clinical setting. Fig. 1 illustrates the categorization of variables in the thalassemia patient dataset, organized into four thematic groups: Demographic & Socioeconomic, Clinical & Treatment, Laboratory Parameters, and Anthropometry & Mental Health.

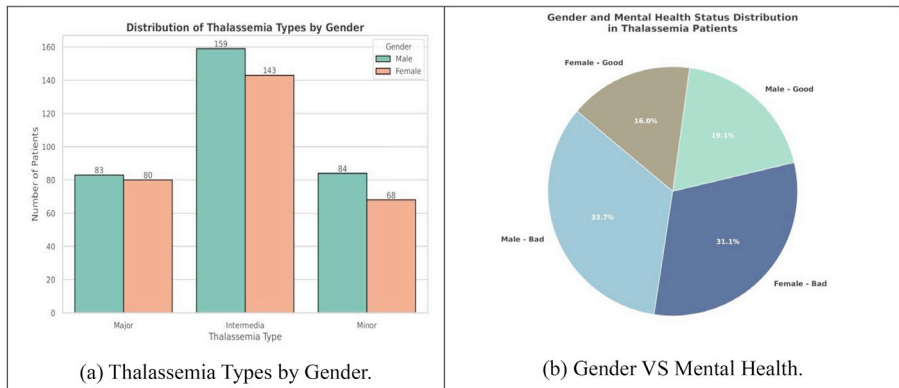


Fig. 2. (a) Gender-based distribution of thalassemia types and (b) mental health status among patients.

The final dataset includes patients aged between 2 and 60 years, with 52.84% identified as male and 47.16% as female. Most patients live in urban areas (58.18%), while 41.82% are from rural areas. In terms of education, nearly half (47.19%) have graduated, followed by secondary (34.49%), primary (10.23%), postgraduate (7.59%), and a small portion (0.49%) with preschool education. Thalassemia intermedia is the most common type (48.95%), followed by major (26.42%) and minor (24.64%). Regarding mental health, 64.83% reported bad mental health, while 35.17% reported good mental health. Fig. 2 shows the gender-based distribution of thalassemia types and mental health status among patients. Fig. 2(a) presents the distribution of thalassemia types by gender, showing that the intermedia type is the most prevalent among both males and females. Fig. 2(b) depicts the distribution of mental health status by gender, revealing that most patients, regardless of gender, experience poor mental health conditions.

4.3. Data preprocessing

To ensure data quality, all entries were double verified by independent reviewers. The variable Self-Reported_Comorbidities includes missing values where patients were unaware of or unable to report comorbid conditions. Additionally, variables such as Education_Level, School/Work_Impact, Iron_Chelation_Therapy, and Chelation_Adherence include entries labeled as 'None' or 'N/A'; however, these do not indicate missing data but rather reflect participants' disagreement, inapplicability, or negative responses. Outlier values were cross verified against original medical records to ensure accuracy. The raw dataset underwent comprehensive preprocessing, which included the removal of duplicate entries and standardization of measurement units (e.g., converting all hemoglobin values to g/dL). Categorical and ordinal variables were recorded as descriptive text labels to improve clarity; users may recode these as needed for statistical or machine learning analyses. To maintain patient confidentiality, all personal identifiers were removed, and everyone was assigned a unique Patient_ID. The final de-identified dataset was exported as a UTF-8 encoded CSV file for broad accessibility.

4.4. Previous studies

Several datasets included in this study have been utilized in prior research exploring various clinical and psychological aspects of thalassemia. In [6], health-related quality of life (HRQoL) among Bangladeshi patients was assessed using the SF-36 questionnaire. Study [7] investigated mental health profiles in Iranian patients with beta-thalassemia major using the GHQ-28 tool. Cognitive and psychological variables in pediatric populations were explored in [8], which adopted structured interviews informed by psychiatric frameworks. In [9], sleep quality and

Table 2
Comparison of thalassemia datasets used in existing studies.

No.	Reference	Repository	Number of Samples	Limitation of Dataset
1	[6]	Mendeley Data	356 patients	Focuses on HRQoL and demographics but lacks detailed clinical data (e.g., genotype, iron overload).
2	[7]	Hospital Collected Data	164 patients	Excludes children under 15 and adults; limits representativeness.
3	[8]	Assiut University Hospital	18–60 (unclear)	Small sample size; unclear sample count; limits generalizability.
4	[9]	Al-Hojaj Clinic	300 patients	Lacks random sampling method.
5	[10]	Self-report Questionnaires	276 patients	Using HADS & SF-36; self-reports can introduce participant bias.
6	[11]	Kaggle	616 samples	Sourced from Kaggle; secondary data; uncertain or inconsistent collection context.
7	[12]	Fatimid Foundation	60 patients	Small sample size; limits statistical power and generalizability.
8	Our Dataset [13]	Mendeley Data	617 patients	Collected from a single center; lacks imaging (e.g., T2/T2* MRI) for assessing organ-specific iron overload.

mental health were analyzed using PSQI and GHQ-28 in a convenience sample of transfusion-dependent thalassemia patients. Study [10], part of a multicenter Thalassemia Longitudinal Cohort, utilized standardized self-report tools (HADS and SF-36) to measure anxiety, depression, and quality of life across diverse demographic groups. In [11], Kaggle-sourced datasets containing CBC and diagnostic features were used to develop machine learning models for thalassemia prediction. Lastly, [12] incorporated a small-scale comparative design to study psychological differences between thalassemia patients and healthy controls, emphasizing the feasibility of such assessments in low-resource clinical settings.

4.5. Highlighting the dataset’s value

Table 2 provides an overview of existing thalassemia datasets, highlighting the number of patients included and key limitations in each study. Most prior datasets primarily emphasize laboratory values or treatment frequency, while our dataset offers a broader scope with 24 variables covering demographic, clinical, socioeconomic, and mental health dimensions. The inclusion of factors like school/work impact, mental health status, and education level alongside key biomarkers enables more holistic, patient-focused research.

Limitations

While this dataset offers valuable insights into thalassemia management in a resource-limited setting, several limitations should be noted. First, data were collected from a single diagnostic centre in Pabna, Bangladesh, which may limit generalizability and introduce selection bias. Second, the lack of advanced imaging, such as T2/T2* MRI, prevents detailed assessment of organ-specific iron overload, restricting evaluation of complications like cardiac or hepatic siderosis. Third, mental health was assessed using a simple self-reported measure (Good” or “Bad”), which lacks the depth of validated psychometric tools and may be influenced by subjective biases or cultural stigma. Similarly, self-reported comorbidities may be incomplete due to patient unawareness. Fourth, thalassemia typing was based on clinical phenotype without genetic confirmation, potentially affecting the precision of epidemiological analyses. Finally, the cross-sectional design over four months precludes longitudinal assessment of disease progression, treatment adherence, or seasonal variation. These limitations underscore the need for multi-centre, genetically informed, and longitudinal studies to complement and expand upon this resource.

Ethics Statement

The study protocol was reviewed and approved by the medical and administrative authorities at LABAID LTD. (Diagnostic), Pabna, Bangladesh. All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments. Participation was voluntary, and no personally identifiable information was collected. All data were anonymized at the point of collection to ensure the confidentiality and privacy of the participants. Informed consent was obtained from all individual participants included in the study. For participants under the age of 18, consent was obtained from a parent or legal guardian. The dataset was created exclusively for academic and research purposes and does not contain any sensitive or clinical identifiers that could be traced back to individual patients.

Credit Author Statement

Shahriar Siddique Ayon: Conceptualization, Methodology, Original draft preparation, **Md. Ebrahim Hossain:** Data curation, Writing, Investigation, **Md. Saef Ullah Miah:** Supervision, Writing- Reviewing and Editing, **B. M. Mredul Arafin:** Data curation, Investigation, **Antara Chowdhury:** Methodology, Writing, **Nuzhat Noor Islam Prova:** Software, Visualization.

Data Availability

The dataset generated and analysed during the current study is publicly available on Mendeley Data at: <https://data.mendeley.com/datasets/8kcdkxmcjw/1>.

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Declaration of Competing Interest

The authors declare no conflict of interest between them or any other parties.

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